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LUBRICANT OIL BEHAVIOR SUBMITTED TO DIFFERENT OPERATING PARAMETERS

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ABSTRACT

Since the second half of the 20^{tth} century it has been great challenges in engines development, with more compact vehicles and the use of new lubricant oil and fuel technologies associated to increased power, performance and consequently the reduction of the emissions levels. By tribology view, science and technology that study the interaction between surfaces in relative motion, it means an increase of specific loads, speeds and temperatures in engine components submitted to friction [(1)].

This work studies the behavior of lubricant oil SAE 15W40 between tribological pairs of three different surfaces treatments technologies against standard ring of gray casting iron. It was used a bench test with block on ring geometry modified and unidirectional sliding movement utilized to evaluate the wear mechanisms, friction and lubrication of the piston ring-cylinder bore of internal combustion engines.

The tribological bench tests showed the variation of structural, interfaces and functions parameters, without the utilization of fuel and consequently not presence of combustion. The results of the lubricant oil analysis were attended through statistics software that determines the main effect and interactions, as well as the evaluation of the worn surface appearance for different operating conditions.

INTRODUCTION

The trends in engine design indicate that tomorrow's engines will demand a great compromise from their lube systems. They will require small package size and low weight, ample pressure for bearing lubrication and ample flow for turbo, efficient piston cooling while maintaining low power consumption from the pump.

The tribological condition of the whole group piston ring-cylinder bore, valve train and engine bearings caused a better comprehension on the different concepts of lubrication modes, during each working cycle, like it's showed in the Figure 1 below in the modified *Stribeck* Diagram [(2)].

In hydrodynamic lubrication, load or pressure is supported essentially by lubricant film, and the contact characteristics are influenced by lubricant properties. In case of boundary lubrication, films adhesion occurs on loaded surfaces. Otherwise in mixed regime, both lubricant film and surfaces support the load and both influence the contact characteristics. In this regime, higher is the load supported by the contacts, higher will be the probabilities for wear development.

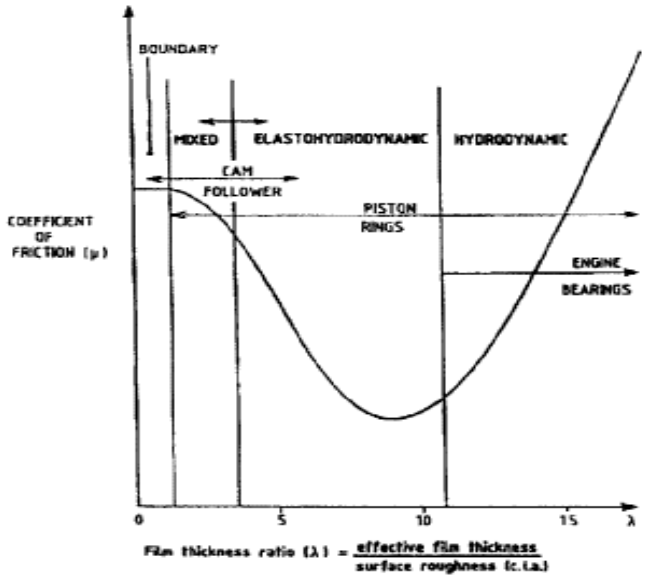


Figure 1 – The modified Stribeck diagram indicates the lubrication mode for each tribological pairs [(2)].

Engine tests have shown that the oil film thickness in the liner-piston contact, depends on critically of the temperature related with the oil viscosity and hence, it's sensitive to the engine speed and torque output conditions.

Decreasing viscosity for improving the fuel economy by reducing the mechanical friction under hydrodynamic conditions will cause simultaneously increased areas of mixed film lubrication regimes. As consequence, the higher wear has to be taken into account. Therefore, a balance between improved fuel economy on one side and increasing wear by lower viscosities on the other side has to be established [3].

The figure 2 illustrate the properties of two mated solids influencing friction and wear during sliding contact. Surface roughness and texturing, which depends on the fabrication method, belong to the micro-geometrical pairing properties. While load, sliding speed, vibration, shape and dimensions of the mated solids belong to the loading and macro-geometrical properties.

The tribochemical properties of the surfaces, such as absorbed atoms, amount and type of surface films, chemical reactivity and thermal and electrical conductivity are not independent of environment properties such as lubricants, humidity, temperature, composition and partial pressure of the gaseous atmosphere. Otherwise, sliding wear is significantly influenced by metallurgical properties [4].

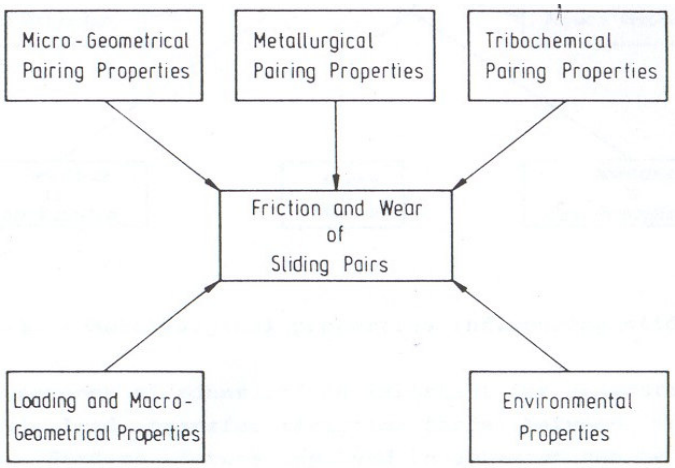


Figure 2 – Properties of two mated solids influencing friction and wear during sliding contact [4].

There are three primary paths for oil consumption: system of power cylinder, engine overhead and PCV. The system that mostly receives the blame for oil consumption and consequently the attention is the power cylinder. The piston ring-cylinder liner plays an important role on the influence of power generating; lubricant consumption and component wear life.

Parallel reductions in the rate of oil consumption and detailed metallurgical modifications became hard to establish a phenomenological link, between soot and wear. Meanwhile, it's commonly believed that good dispersion control is a prerequisite of wear performance [5] [6]

Lubrication is one of the most important demands, and with fundamental influence on wear mechanisms. If a continuous oil film is present, the friction wear will be neglected. However, with the cracking of oil film, the several wear will occur in TDC region, due to speed inversion of the pair ring-piston, causing a depression and polishing of this region, mostly observed in Diesel engines.

Bench test utilizing Electric motor driven engine, promotes lower load, due to combustion absence and consequently lower wear along piston course and also in the TDC region, compared to dynamometer test. It was observed typical wear in bench test, with the presence lube residues (detected by EDX analysis) plastic deformation and scratching on both ring and bore worn surfaces [7].

Some contaminations are difficult to detect as following viscosity trends in the oil may indirectly monitor particulate. The table I shows some techniques to analysis of oil lube particles.

Table I – Summary of the particle characterization techniques [8].

Technique	Particle Size Range	Particle Types
Spectroscopy	Angstroms to about 2-3 micrometers	Wear metals, silicon and soot
Screen Blockage	5 and greater micrometers	No differentiation
Ferrography, Direct and Analytical	1 and greater micrometers	Iron wear debris

As lubricating oil deteriorates with use, contaminants in the oil build up over time. These contaminants include both chemical constituents such as oil oxidation products and acids due to combustion gases and particulate constituents such as dust and dirt from external sources.

Organic material may be implicated in filter plugging and ring and liner deposits. This organic plugging of filters is one of the most important limitations on service intervals when the restriction across the filters becomes too high. This can result in bypassing contaminants and accelerated wear. Sources of internal inorganic contamination are usually from core sand and also residuals from the casting process, which are not completely washed away before engine assembly. This sand is generally flushed from the system after the first several oil drains, but contributes to the high rate of wear during the break-in of a new engine [9].

LUBE OIL BEHAVIOR DURING BENCH TEST

Bench test are generally performed through severe conditions compared to engine operating conditions. The test conditions and contact geometry are quite different from real situation, but there are advantages regarding to time and tribological variable monitoring.

Initially the bench test condition were chosen close to engine operating conditions at TDC (mixed lubrication mode), i.e., high temperature, load produced by combustion pressure and low velocity like the following condition at 100°C, 13 MPa (~ 334N) and 0,5 m/sec. The other test conditions were chosen increasing or decreasing the initial values depending on the machine capacity.

The bench tests were performed in Falex equipment, with block on ring modified geometry, to fix a piston ring segment with approximately 10mm, tested against a standard test ring of perlitic grey cast iron, with mean hardness of 98HRB, diameter of 35mm and grinded with roughness of 0,6µm RA. The scheme of Figure 3, shows the equipment and the assembly condition of the ring segment at test block partially lubricated (counterpart semi-bath). The lubricant used was the SAE 15W40 – API CE, category that changed the market in multiviscous class. The physic-chemic characteristics were analyzed before the tests, and the results are described below:

TAN	= 2,15 mg KOH/g
TBN	= 7,39 mg KOH/g
Viscosity to 40°C	= 102,9 cSt
Flash point	= 216°C

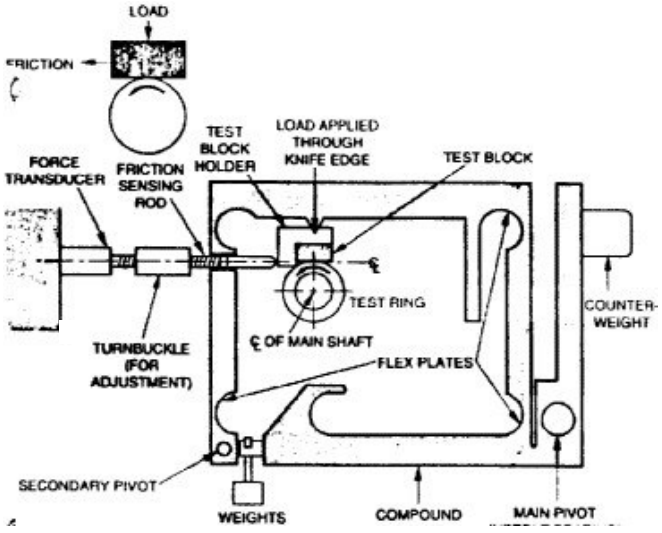


Figure 3 – Assembly Scheme of Falex equipment modified for bench test

The tribological bench test allows the variation of structural, interface and function parameters as indicated in table II.

TABLE II - BENCH TESTS PARAMETERS		
Structural	Interface	Functional
Geometrical* Surface treatments	Lube oil 15W40*	Load Speed Oil Temperature
*similar characteristics for all experiments		

The experiment planning was divided in two factorial experiments with L12 arrangement and the parameters are indicated in Table III. (5 repeat to each experiment)

1) Load and speed (2^2) and material (3^1) temperature 100°C

2) Load and temperature (2^2) and material (3^1) speed 273 rpm

Table III – Planning of wear bench tests

Test	Load (N)	Speed (rpm)	Temperature (°C)	Surface Treatment
1	334	273	25	Plasma Spray
2	334	273	25	Gas Nitriding
3	334	273	25	Chromium Plating
4	334	273	100	Plasma Spray
5	334	273	100	Gas Nitriding
6	334	273	100	Chromium Plating
7	334	820	100	Plasma Spray
8	334	820	100	Gas Nitriding
9	334	820	100	Chromium Plating
10	667	273	25	Plasma Spray
11	667	273	25	Gas Nitriding
12	667	273	25	Chromium Plating
13	667	273	100	Plasma Spray
14	667	273	100	Gas Nitriding
15	667	273	100	Chromium Plating
16	667	820	100	Plasma Spray
17	667	820	100	Gas Nitriding
18	667	820	100	Chromium Plating

Load – Assuming that the normal load, generated by contact pressure is the sum of load variations during a working cycle, when the thickness of oil film is lower then the minimum thickness for hydrodynamic lubrication. Then it has the border condition to establish contact between two surfaces, and consequently the wear.

Speed – This factor is important to determine changes in contact surfaces. On superficial oxides formation, as the sliding speed increases, lower is the time for oxide formation, improving the superficial temperature, helping the dissipation of weak protection films presents on the surface, or also improving the oxidation reaction depending of other conditions like lubricate quality and load regimes.

Temperature – In determined conditions, occurs the dissipation of protection films, however accelerating oxides formation and protecting the contact surface. In lower temperatures, there is no quick formation of oxides, then no such film dissipation. Although if the temperature is ideal for film dissipation, so without a quick enough oxide formation, the surface would be unprotected and a stronger worn should occur by the adhesion mechanism.

Materials – The top piston ring is applied for high performance application so the surface treatments of Plasma Spray (Mo) achieving 400HV, Gas Nitriding (Nitret) achieving 1100HV and Chromium Plating (Cr comp) achieving 980 HV are used, all containing hard particles dissipated.

In the analyses to verify lubricant oil behavior were utilized Physical-chemical method to analyses viscosity at 40ºC and 100ºC, viscosity index and TBN/TAN. For metallic particles as Zn and Fe was used optical absorption spectrophotometer. Oxidation content was measured by infrared equipment.

To determine the wear mechanisms, like material adhesion, plastic deformation hard particles scratching or even surface phenomenon, was used electronic scanning microscopy with the deep focus resource and microanalysis system (EDX).

The contact ellipse area at the end of the tribological test indicates the wear results of surface treatments of each piston ring segment. The figure 4 illustrates the worn aspect (above piston ring) and the respective standard test ring after running a distance of 5,5km during the sliding in bench test.

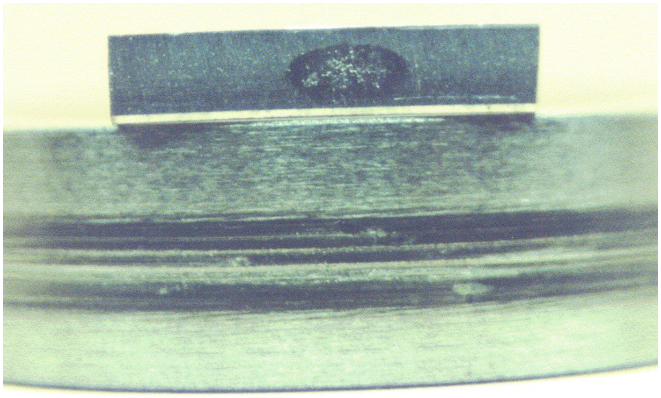


Figure 4 – Appearance of surface treatment and standard ring after sliding distance of 5,5 km.

STATISTICS ANALYSIS OF RESULTS

The software MINITAB 2000 version XIV provides various options to data statistical analysis. In this work was used the multiple comparison method with a confidence interval of 90%, and experiment planning techniques to determine the most significant factors.

MAIN EFFECTS AND INTERATIONS

Viscosity to 40℃ – The figure 5 show the temperature increasing, followed by the oxidation, and consequently higher viscosity index, mostly when using lower loads, the same behavior was noticed to 100℃ viscosity. The oxidation reaction can be helped by other parameters such as lubricant oil characteristics and load condition.

Viscosity Index – The viscosity variation is less noticed when using a superficial treatment with the Plasma Spray process, for the Gas Nitriding and Chromium Plating is observed a larger variation for different load conditions, as showed in Figure 6.

Figure 5

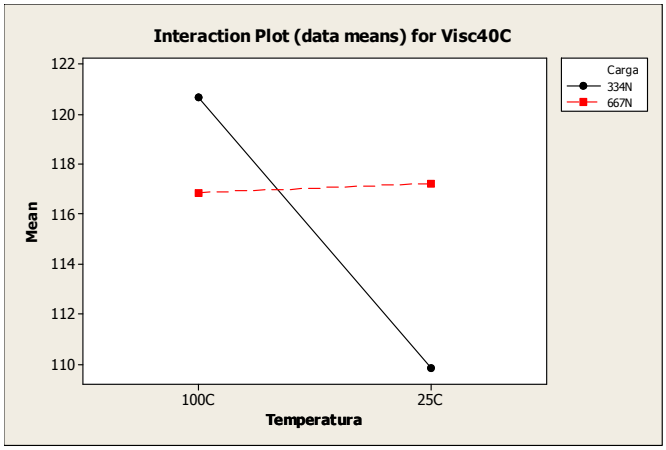
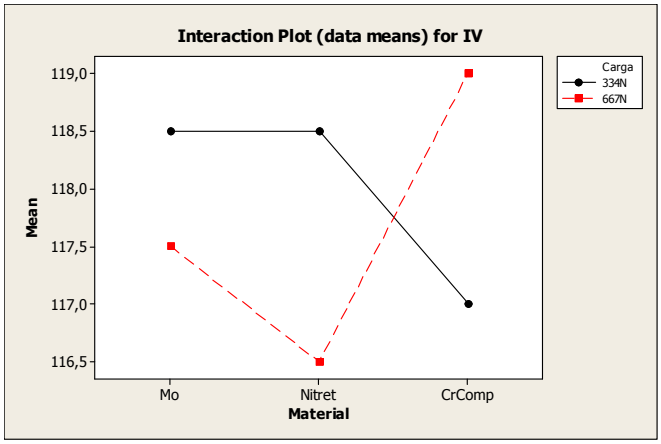
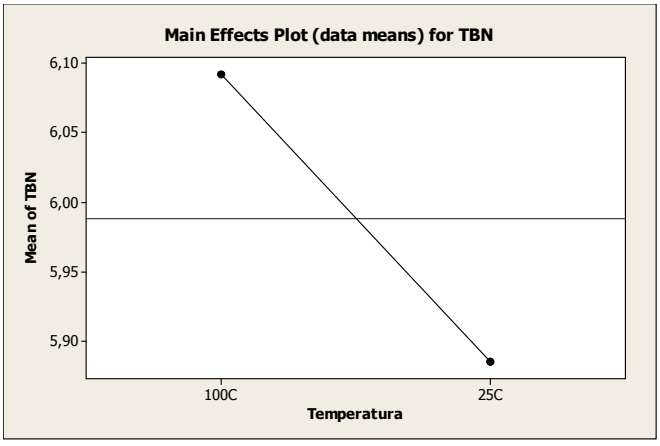


Figure 6



TBN (Total Base Number) – Indicates the effective alkaline reserve of the lubricant oil. In Figure 7 is observed a decrease of TBN values, when tested in lower temperatures, most probable due to air humidity to this condition.

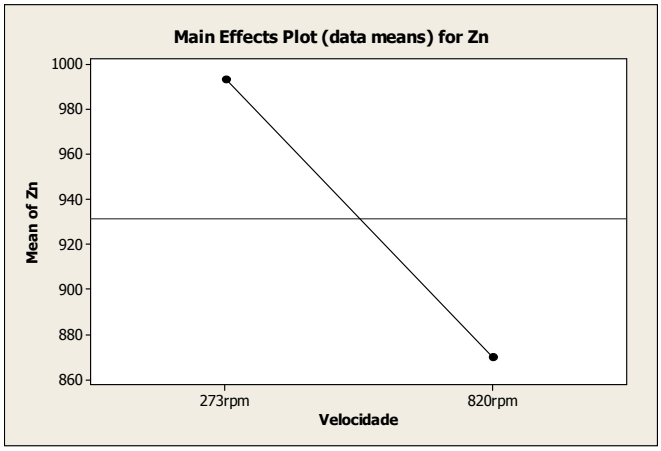
Figure 7



Zinc - This element is presence in anti-wear and oxidation inhibitor chemical know as ZDP (zinc dithophosphate) of the lubricant oil. In Figure 8 is

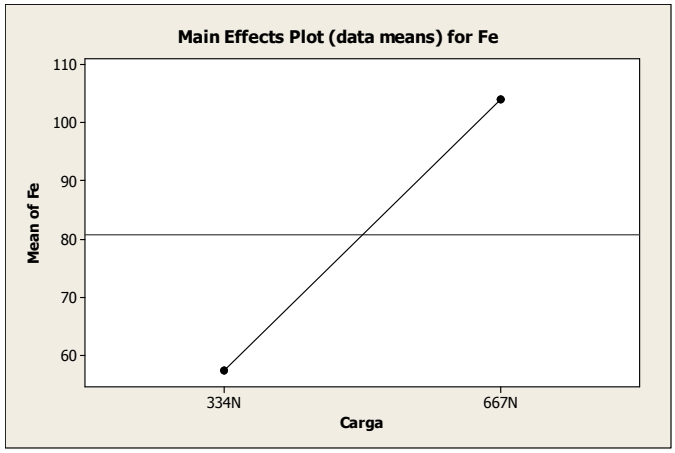
observed with the speed increasing, lower are the Zinc values, contained in the oxide film formation between the contact surfaces (lubricant residual).

Figure 8



Iron – Present mostly on cast iron of the standard test ring and also in the Gas Nitriding superficial treatment. The figure 9 indicates that increasing the load, higher is the presence of Iron in the lubricant oil, due action of some wear mechanism associated to temperature increase the quantity of this element is even higher.

Figure 9



Scratching Level (Abrasion) – The Figure 10 shows that speed increment results in soft scratching level increased, this is most evident in the Gas Nitriding superficial treatment. The Figure 11 shows the scratching level on the worn region of the Gas Nitriding surface.

Figure 10

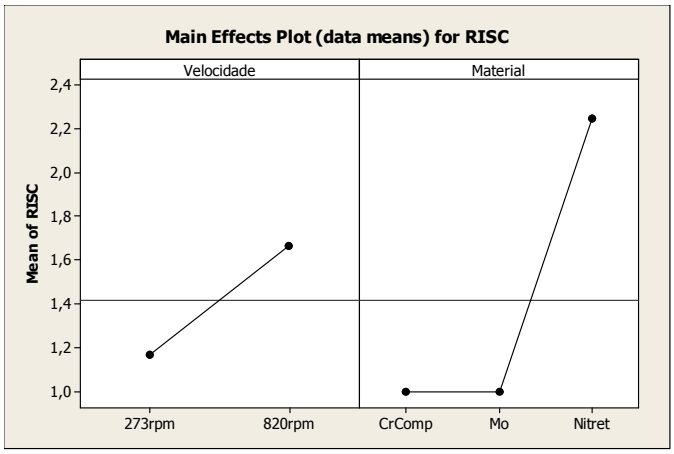
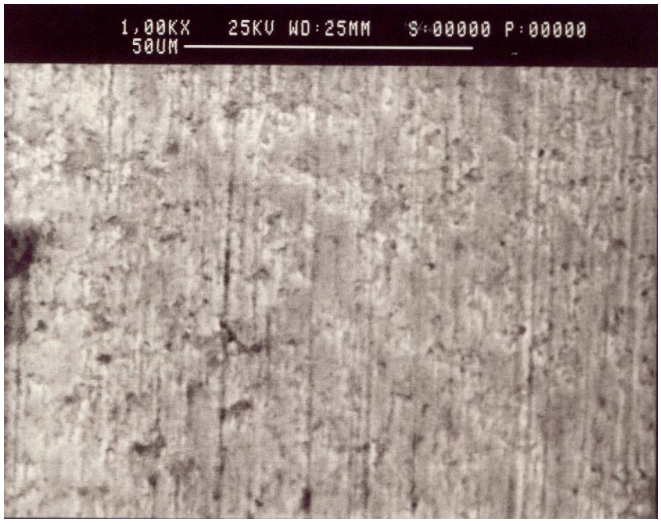


Figure 11



Lubricant Residual – The Figure 12 shows that superficial treatment Chromium Plating reveals higher quantity of oil lubricant residue (protective films), in comparison to other technologies. The Figure 13 shows the lubricant residue in the worn region of the surface treatment Chromium Plating. EDX analysis of lubricant residual in the worn bore surface shows elements as Zn, Ca, Mg from oil additives.

Figure 12

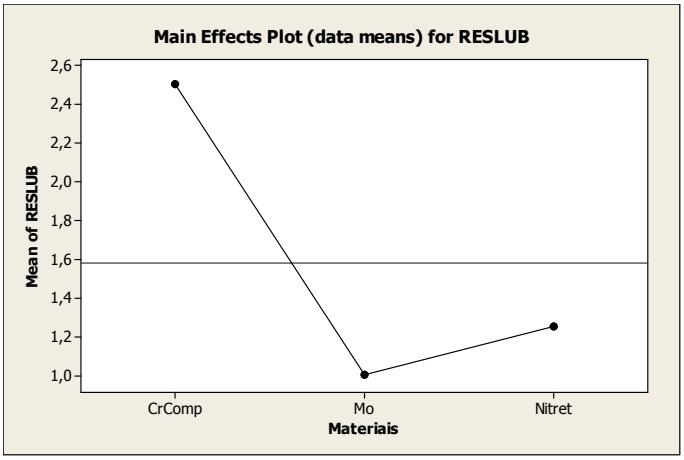
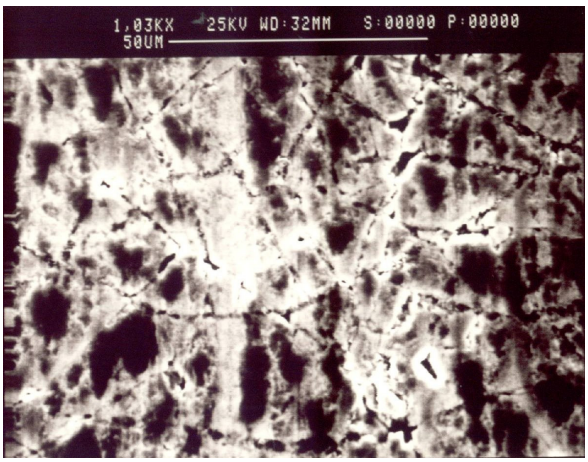


Figure 13



Between the surface treatments of the piston ring segment, the higher worn was observed in the mode 13, using 667N, 273rpm e 100°C with Plasma spray coating, while the modes 5 and 6 showed lower worn with surface treatments of Chromium plating and Gas Nitriding respectively, both in 334N, 273rpm and 100°C condition, that indicates a large influence in load parameters and oil lubricant temperature. The figures 14 and 15 show the effects relative to wear for each technology of superficial treatment.

Figure 14

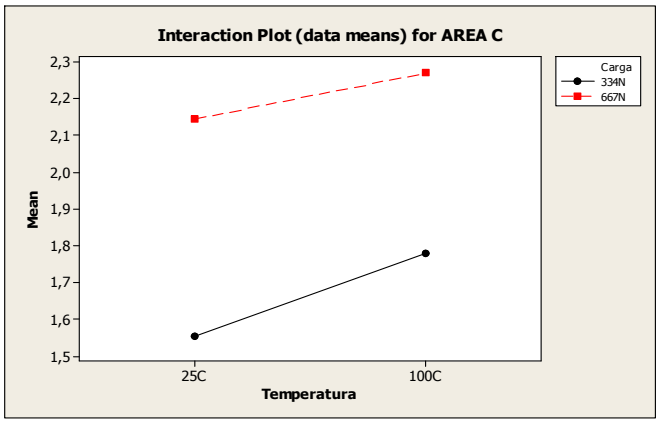
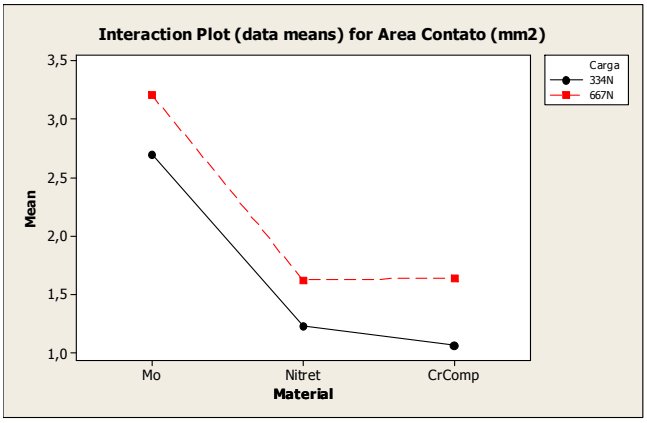


Figure 15



CONCLUSIONS

The bench tests indicated variations on lubricant oil behavior and also in wear on both technologies of superficial treatment, even with the combustion absence could be observed effects and significant interactions due to operating conditions.

The combustion provides the tribochemical mechanism allied to the corrosive and oxidative wear determined by load factors and sliding speed, with the variation of lubrication regime mostly in the TDC region.

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